Health Technology Ecosystem

Centers for Medicare & Medicaid Services (CMS), Assistant Secretary for Technology Policy (ASTP)/Office of the National Coordinator for Health Information Technology (ONC) (collectively, ASTP/ONC), Department of Health and Human Services (HHS).

Request for Information (RFI): CMS-0042-NC

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Being Involved & Being Responsible

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CMMI DEV | ML3 APPRAISED Appraisal # 64879 | Exp. Sep 12, 2026

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Cover Page

iAdeptive Technologies appreciates the opportunity to provide CMS and ASTP/ONC input on advancing digital health and interoperability. We commend the agencies' ongoing efforts to foster a secure, patient-centered data ecosystem that empowers individuals and enables innovation.

iAdeptive Technologies is a digital transformation and analytics partner with deep experience in federal healthcare IT. We support CMS in modernizing enterprise systems and driving value-based care transformation through data-driven innovation. Our work includes leading initiatives for Hospital Quality Reporting (HQR) and End-Stage Renal Disease Quality Reporting System (EQRS) and implementing a modern Contract Management solution (ODLRA) that integrates advanced analytics, automation, and audit capabilities. Across these programs, we have built AI-driven insights platforms, established secure, scalable AWS-based data lakes, and delivered FHIR-compatible APIs to facilitate interoperability across payers, providers, and regulatory agencies.

Given the public nature of this RFI, our detailed architectural approaches and proprietary AI models are not disclosed here. However, we welcome the opportunity to provide in-depth briefings or demonstrations in a secure, closed-door setting.

Please refer to the **Policy Innovation Memo** below for forward-looking perspectives on reframing interoperability in the age of AI and recommendations for a centralized patient access model.

This section's brief summarizes iAdeptive's response to the CMS and ASTP/ONC Request for Information (RFI) regarding digital health technology and data interoperability. As an experienced HHS/CMS vendor of AWS-based AI/data lake solutions, we offer actionable insights on streamlining health IT infrastructure, enhancing patient access, and reducing provider burden.

Key Themes & Recommendations

- Simplify developer experience with better APIs, sandbox realism, and open documentation (TD-1, TD-2).
- Enable Medicare beneficiaries to access all health data via a centralized CMS-operated portal, modeled after IRS.gov (TD-2, TD-13).
- Promote AI-native ingestion pipelines to reduce reliance on rigid FHIR standards (TD-7, TD-9).
- Encourage bulk EHR integration for quality reporting, reducing submitter friction (TD-9e, TD-15).
- Leverage identity frameworks like Login.gov while ensuring digital equity (TD-3).

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- Advance public health through transparent price APIs and de-identified bulk FHIR support (TD-15, TD-19).

Next Steps

We welcome the opportunity to present our AI architecture, bulk FHIR solutions, and patient access model in a closed-door session. Our technical teams are ready to support CMS' shift toward value-based care through intelligent, patient-centered data platforms.

E. Technology Vendors, Data Providers, and Networks

TD-1. How CMS Can Stimulate Developer Interest (Short- & Long-Term)

Short-Term Steps (Next 2 Years)

1. **Open-Source CMS APIs and Encourage Community Participation**
    - CMS should open source the Blue Button, DPC, and BCDA APIs and create a Developer Community Edition that enables external contributors to propose enhancements, report issues, and extend capabilities.
    - A structured community governance model—similar to successful open-source frameworks like VA Lighthouse or Fast Healthcare Interoperability Resources (FHIR)—can foster innovation, transparency, and rapid development of new features relevant to patients and caregivers.
2. **Improve Sandbox & Documentation** – Add realistic synthetic data, clarify migration paths, and resolve link issues.
3. **Fix Blue Button Data Quality** – Address missing/mismatched IDs and NDC codes.
4. **Faster Support Responses** – Allocate more CMS staff to assist dev forums (e.g., Google Group).
5. **Clear Security Guidelines** – Provide HIPAA-compliant implementation best practices.

Long-Term (2+ years):

1. **Host Hackathons/Developer Events** – Promote collaboration and community building.
2. **Launch Grants & Prizes** – Fund apps improving patient engagement or care delivery.
3. **Promote User-Centered Design** – Align APIs with real-world patient and caregiver needs.
4. **Simplify Identity Access** – Make Login.gov universal for all CMS developer apps.

TD-2. CMS Data Expansion & API Innovation

a. Most Valuable Additional Data Sources:

- Medicaid, Medicare Advantage, and Marketplace data
- Provider performance & APM alignment data
- MyChart-style EHR APIs
- Social risk factors (housing, food insecurity)

b. Key Barriers Today:

- Lack of beneficiary OAuth2 consent model
- HIPAA limitations for non-covered entities
- Fragmented tech across Blue Button, BCDA, DPC
- MA and Medicaid controlled by external orgs
- Minimal sandbox realism and proxy access

c. Additional APIs CMS Should Prioritize:

- Caregiver proxy API
- Medicare card & eligibility validation
- Real-time APM attribution
- Fraud flagging & audit metadata
- Feedback API for patient & developer input
- Metrics API to track usage by feature

TD-3. Digital Identity: Challenges, Impacts, and Recommendations

a. Challenges & Benefits

- **Benefits**: Stronger security, reduced fraud, and interoperability across platforms.
- **Challenges**:
  - Many users (especially elderly, rural, low-income) lack the tech literacy or infrastructure to navigate centralized identity systems.
  - Small orgs may not have the resources to integrate NIST 800-63-3-compliant solutions.

b. Impact of Credentialing (Login.gov, ID.me, CLEAR):

- **Positive**: Boosts trust and compliance with HIPAA; reuses existing proofing from HealthCare.gov.
- **Negative**: It could widen the digital divide unless assisted identity proofing (e.g., in-person verification or call-in support) is provided.

c. Mandating OpenID Connect:

- Promotes modern, secure identity federation.

- However, this incurs upfront costs for providers and developers—grants or technical support may be needed.
- CMS should **avoid vendor lock-in** and encourage multiple CSP (Credential Service Provider) options to ensure flexibility and equity.

TD-4. Encouraging Open, Standards-Based APIs

Key Recommendations:

1. **Publish Real-World Case Studies**
   Demonstrate how FHIR-based APIs improved Medicare populations' outcomes, access, and innovation.
2. **Foster Developer Ecosystem**
   Fund FHIR sandboxes, launch challenges, and support third-party app marketplaces aligned to CMS APIs.
3. **Align Incentives to Outcomes**
   Tie provider reimbursements or quality scores to their adoption of open APIs for data access and interoperability.

TD-4. Encouraging Open, Standards-Based APIs

- **Showcase Success Stories**: Publish patient-focused case studies where open APIs (e.g., FHIR) improved care coordination or access.
- **Build a Developer Community**: Create incentives, challenges, and sandboxes to support third-party app development using CMS APIs.
- **Tie to Incentives**: Reward providers who enable API-based data sharing with quality bonuses or reduced reporting burden.

TD-5. Nationwide Provider Directory of FHIR Endpoints

- **Value**: Improves routing of FHIR-based requests for patients, payers, and providers.
- **Recommendation**: CMS should publish and maintain a free, standardized, searchable directory integrated with NPPES.
- **User Access**: Keep it free and public modeled after Care Compare or AMA Finder.

TD-6. TEFCA's Role + Recommendation for Dual-Path Interoperability

- **Current Function**: TEFCA provides a legal and governance framework for nationwide interoperability across networks and stakeholders.
- **Opportunity**: TEFCA aims to unify data exchange via FHIR—but current healthcare IT still relies heavily on legacy standards like HL7 v2 and CCDs.

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iAdeptive Recommendation: Dual Approach

1. **FHIR Path (for Compatibility)**
   Continue maturing FHIR as the nationwide standard for structured, standards-based exchange. TEFCA can promote adoption and compliance.
2. **AI-Enabled Path (for Flexibility and Equity)**
   iAdeptive recommends CMS invest in **AI-based ingestion and auto-domain identification** tools that allow:
   - Submission of raw or semi-structured data (CSV, flat files, PDFs)
   - Automated data classification (e.g., labs vs. claims vs. diagnosis)
   - Domain tagging via machine learning
   - Reduced development burden on providers, payers, and intermediaries

This **lowers the barrier for small orgs and safety-net providers**, accelerating participation and data sharing without complete technical alignment to FHIR.

TD-7. USCDI: Progress, Limitations & Dual-Path Recommendation

Progress:

- USCDI has improved baseline interoperability and EHR data standardization.
- Supports structured exchange for core data elements across providers and systems.

Limitations:

- Misses richer context from unstructured data (e.g., lab timestamps, disability status).
- May not keep pace with real-world data complexity and AI-enabled workflows.

iAdeptive's Dual Approach

1. **USCDI Path (Standards-Based Compliance)**
   Continue evolving USCDI to expand domains (e.g., behavioral health, SDOH, functional status) while maintaining regulatory alignment.
2. **AI-Native Path (Flexible Data Enablement)**
   Certify CMS-approved ingestion engines that:
   - Accept semi-structured inputs (PDFs, CSVs, raw notes)
   - Use ML/NLP to classify, map, and extract relevant fields
   - Surface-structured elements aligned to USCDI when possible

This allows broader participation across data submitters without requiring full FHIR/USCDI conformance upfront.

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TD-8. Effective Certification Standards

iAdeptive supports prioritizing:

- **API certification (FHIR/SMART)**
- **USCDI alignment**
- **OAuth2-based access controls**

These standards directly support 21st Century Cures Act goals—especially around patient access, third-party app integration, and data security.

Certification should reflect not only compliance, but also enablement of real-world use.

TD-9. Rethinking Certification for APIs

a. Benefits of Prioritizing APIs:

- Easier integration and modular upgrades
- Promotes open data access
- Supports faster innovation cycles

b. Drawbacks:

- API focus may under-specify core clinical workflows
- Smaller providers may lack API integration capacity
- APIs without verified data logic risk poor reliability

c. iAdeptive Suggests:

- Certify **AI-based unstructured data handlers** to extract and validate text-based/faxed/legacy formats
- Provide interim certification for **hybrid systems** that support both FHIR APIs and NLP-enhanced ingest paths

TD-10. Enabling API Access Without Special Effort

Key Recommendations:

- Require **uniform FHIR implementation guides** across certified vendors
- Mandate **EHI and full USCDI exposure** through certified APIs
- Ensure SBX environments reflect **real-world data complexity**
- Establish a **transparent complaint process** for blocked or non-functional API implementations

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Access without special effort means minimizing both technical friction and regulatory ambiguity.

TD-11. EHI Export Standardization

a. Standardized API Requirements:
Yes. Current flexibility leads to inconsistent implementations. CMS/ONC should mandate FHIR-based EHI exports with defined minimum data sets and consistent formats.

b. Workflow Improvements Needed:

- Simplify patient-level export flows
- Automate population-level exports with filters (e.g., by condition, time period)

c. Policy Support Suggestions:

- Require export endpoints to be accessible to both patients and authorized third parties
- Tie certification to demonstrated ease-of-use and data completeness

TD-12. Non-CMS Data Source Endorsements

Yes—CMS should endorse high-quality external data networks, using criteria such as:

- **Completeness & freshness of clinical data**
- **Participation in CMS-aligned programs (e.g., EQRS, HQR)**
- **Standards adoption (e.g., HL7 FHIR)**
- **Patient access and consent support**

Examples to prioritize:

- Dialysis networks (e.g., EQRS)
- HIEs aggregating cross-provider longitudinal data

TD-13. Full EHI API Access: Opportunities & Tradeoffs

Opportunities:

- Richer AI and analytics across the full care continuum
- Summarization, care plan automation, fraud detection

Obstacles:

- Privacy risks and consent complexity
- App developer readiness (especially for smaller orgs)

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USCDI vs. Full EHI Tradeoff:

- USCDI is leaner but less expressive
- Full EHI enables richer models, but requires stronger safeguards

iAdeptive supports a dual-tiered model: USCDI for minimum interoperability, full EHI via AI/NLP pipelines for advanced insights.

TD-14. Networks & FHIR API Usage

Note: As a technology vendor, iAdeptive does not operate a FHIR-connected data network but recommends:

- Encouraging **registry-based endpoints** searchable by NPI/org ID
- Promoting **network interconnectivity** via TEFCA and open API frameworks
- Supporting **cross-network agreements** for seamless patient record exchange

TD-15. Bulk FHIR APIs

a. Advantages of Expansion:

- Scales data access for ACOs, researchers, and app developers
- Supports AI/ML model training with real-world data
- Reduces reporting burden via structured, batched submissions (e.g., for EQRS, HQR)

b. Disadvantages/Concerns:

- Risk of data misuse or re-identification
- Requires stronger authentication, rate-limiting, and consent controls
- Misinterpretation risk without adequate metadata or documentation

iAdeptive encourages CMS to certify Bulk FHIR APIs and align them with reporting programs (e.g., CCSQ, ESRD).

TD-16. Point-to-Point vs. Shared Infrastructure

Shared Infrastructure Benefits:

- Reduced duplication of APIs
- Easier developer onboarding
- Scalable, OAUTH-backed access model (e.g., Blue Button)

Improvements Needed:

- **Unify APIs** (Blue Button, BCDA, DPC) under one common stack
- Expand coverage to **Medicaid and Medicare Advantage**
- Allow reciprocal access between providers, payers, and apps

CMS should lead consolidation of APIs into a single trust-based infrastructure.

TD-17. Supporting Data-Sharing Network Viability

CMS and ONC should:

- Provide **financial support** (grants/incentives) for smaller orgs
- Establish **usage benchmarks** tied to real-world outcomes
- Require participation as part of value-based care or payment models
- Promote **community governance models** to sustain trusted exchange

TD-18. Information Blocking Practices

Examples of Information Blocking:

- Repeated API failures or token revocations without explanation
- Rate limiting that prevents meaningful use
- Non-responsiveness to FHIR endpoint onboarding requests

Recommended Policies:

- Tie participation in CMS programs to **measurable transparency scores**
- Establish **reporting mechanisms** and corrective timelines
- Penalize persistent non-cooperation through VBC penalties

Actors Needing More Oversight:

- Providers still face weak disincentives—rules should be enforced equally

TD-19. Price Transparency

c. Current Shortcomings:

- Lack of real-time, service-specific cost estimates
- No predictive out-of-pocket cost tools
- Limited access to negotiated rates pre-service

d. High-Impact Workflows:

- Pre-procedure planning

- Chronic care budgeting
- Claims reconciliation and appeals

e. Suggested Improvements:

- Build FHIR-based cost lookup APIs
- Create patient-facing tools like Medicare "Cost Explorer"
- Enable patients and providers to **flag price anomalies or fraud**

True transparency requires real-time pricing at the decision point—not just retrospective claims access.

Conclusion

iAdeptive Technologies values this opportunity to contribute to the evolving dialogue on digital health transformation. We strongly support CMS and ASTP/ONC's commitment to patient-centered innovation, reduced provider burden, and scalable interoperability. We recommend a dual-path approach—supporting both FHIR-based standards and AI-native data ingestion models—to meet the diverse needs of the ecosystem. Given the public nature of this RFI, we have limited our disclosures here, but we would welcome a dedicated meeting to demonstrate our solutions and discuss detailed technical pathways in a secure, collaborative setting.